

Proving Equations Using Identities (Homework)

Prove that each equation is true by converting the left side to the right.

$$\textcircled{1} \sec A \sin A \cos A = \frac{1}{\csc A}$$

$$\textcircled{2} \sin^2 \theta - 2 + \cos^2 \theta = -1$$

$$\textcircled{3} \sec^2 y + 51 = \tan^2 y + 52$$

$$\textcircled{4} \tan w (\cot w + \tan w) = \sec^2 w$$

$$\textcircled{5} \cos^2 B (\sec^2 B - 1) = \sin^2 B$$

$$\textcircled{6} \frac{\sin^2 \beta - \cos^2 \beta}{\sin \beta \cos \beta} = \tan \beta - \cot \beta$$

$$\textcircled{7} \cos x - \cos x \sin^2 x = \cos^3 x$$

$$\textcircled{8} \frac{1 - \tan^2 \alpha}{1 + \tan \alpha} = 1 - \frac{\sin \alpha}{\cos \alpha}$$

$$\textcircled{9} \frac{\cot^2 \theta - 2 \cot \theta - 15}{\cot \theta - 5} = \frac{1}{\tan \theta} + 3$$

$$\textcircled{10} \frac{\csc^2 A - \sec^2 A}{\csc A + \sec A} = \frac{1}{\sin A} - \frac{1}{\cos A}$$

$$\textcircled{11} \csc B - \cot B \cos B = \sin B$$

$$\textcircled{12} \tan \theta + \sec \theta = \frac{\sin \theta + 1}{\cos \theta}$$

$$\textcircled{13} \cot w + \frac{\sin w}{\cos w - 1} = \frac{1}{\sin w}$$

$$\textcircled{14} \frac{1}{1 - \cot A} + \frac{1}{1 - \tan A} = 1$$

$$\textcircled{15} \frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} = 2 \sec^2 x$$

Homework Answers

* There is more than one way to prove each equation below. Each solution demonstrates only one.

$$\textcircled{1} \sec A \sin A \cos A = \frac{1}{\csc A} \quad \textcircled{2} \sin^2 \theta - 2 + \cos^2 \theta = -1$$

$$\frac{1}{\cancel{\cos A}} \sin A \cancel{\cos A} =$$

$$\sin A =$$

$$\frac{1}{\csc A} =$$

$$\sin^2 \theta + \cos^2 \theta - 2 =$$

$$1 - 2 =$$

$$-1 =$$

$$\textcircled{3} \sec^2 y + 51 = \tan^2 y + 52$$

$$\tan^2 y + 1 + 51 =$$

$$\tan^2 y + 52 =$$

$$\textcircled{4} \tan w (\cot w + \tan w) = \sec^2 w$$

$$\tan w \cot w + \tan^2 w =$$

$$\tan w \frac{1}{\tan w} + \tan^2 w =$$

$$1 + \tan^2 w =$$

$$\sec^2 w =$$

$$\textcircled{5} \cos^2 B (\sec^2 B - 1) = \sin^2 B$$

$$\cos^2 B \sec^2 B - \cos^2 B =$$

$$\cos^2 B \frac{1}{\cos^2 B} - \cos^2 B =$$

$$1 - \cos^2 B =$$

$$\sin^2 B + \cos^2 B - \cos^2 B =$$

$$\sin^2 B =$$

$$\textcircled{6} \frac{\sin^2 \beta - \cos^2 \beta}{\sin \beta \cos \beta} = \tan \beta - \cot \beta$$

$$\frac{\sin^2 \beta}{\sin \beta \cos \beta} - \frac{\cos^2 \beta}{\sin \beta \cos \beta} =$$

$$\frac{\sin \beta}{\cos \beta} - \frac{\cos \beta}{\sin \beta} =$$

$$\tan \beta - \cot \beta =$$

$$\textcircled{7} \cos x - \cos x \sin^2 x = \cos^3 x$$

$$\cos x (1 - \sin^2 x) =$$

$$\cos x (\sin^2 x + \cos^2 x - \sin^2 x) =$$

$$\cos x \cdot \cos^2 x =$$

$$\cos^3 x =$$


$$\textcircled{8} \frac{1 - \tan^2 \alpha}{1 + \tan \alpha} = 1 - \frac{\sin \alpha}{\cos \alpha}$$

$$\frac{(1 - \tan \alpha)(1 + \tan \alpha)}{1 + \tan \alpha} =$$

$$1 - \tan \alpha =$$

$$1 - \frac{\sin \alpha}{\cos \alpha} =$$


$$\textcircled{9} \frac{\cot^2 \theta - 2\cot \theta - 15}{\cot \theta - 5} = \frac{1}{\tan \theta} + 3$$

$$\frac{(\cot \theta - 5)(\cot \theta + 3)}{\cot \theta - 5} =$$

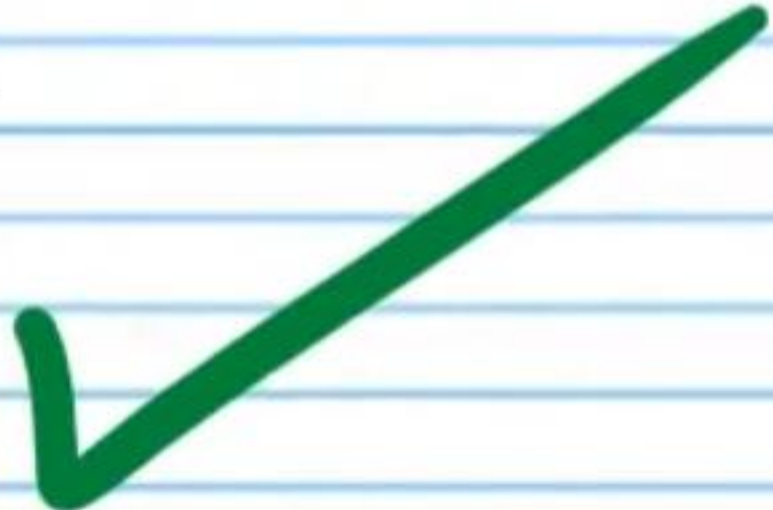
$$\cot \theta + 3 =$$

$$\frac{1}{\tan \theta} + 3 =$$


$$\textcircled{10} \frac{\csc^2 A - \sec^2 A}{\csc A + \sec A} = \frac{1}{\sin A} - \frac{1}{\cos A}$$

$$\frac{(\csc A - \sec A)(\csc A + \sec A)}{\csc A + \sec A} =$$

$$\csc A - \sec A =$$

$$\frac{1}{\sin A} - \frac{1}{\cos A} =$$


$$\textcircled{11} \csc B - \cot B \cos B = \sin B$$

$$\frac{1}{\sin B} - \frac{\cos B}{\sin B} \cos B =$$

$$\frac{1}{\sin B} - \frac{\cos^2 B}{\sin B} =$$

$$\frac{1 - \cos^2 B}{\sin B} =$$

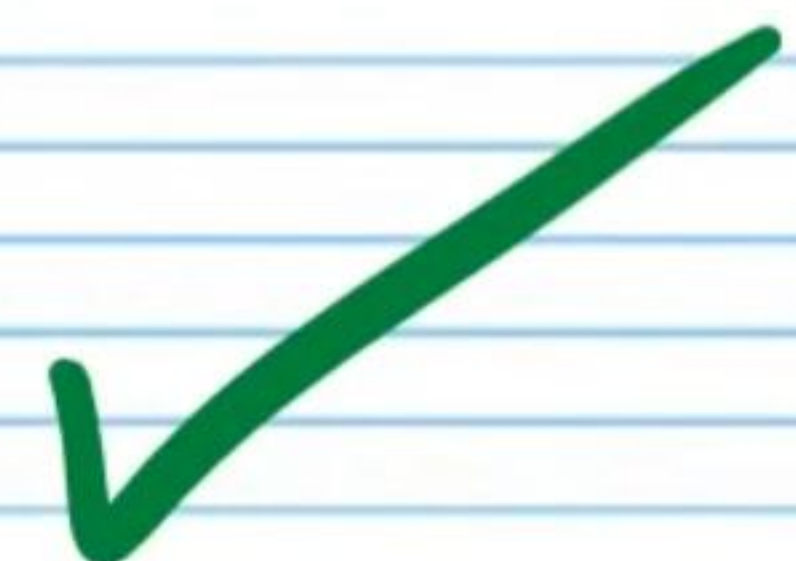
$$\frac{\sin^2 B + \cos^2 B - \cos^2 B}{\sin B} =$$

$$\frac{\sin^2 B}{\sin B} =$$

$$\sin B =$$


$$\textcircled{12} \tan \theta + \sec \theta = \frac{\sin \theta + 1}{\cos \theta}$$

$$\frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta} =$$

$$\frac{\sin \theta + 1}{\cos \theta} =$$


$$\textcircled{13} \cot W + \frac{\sin W}{\cos W + 1} = \frac{1}{\sin W}$$

$$\frac{\cos W}{\sin W} + \frac{\sin W}{\cos W + 1} =$$

$$\frac{\cos W (\cos W + 1)}{\sin W (\cos W + 1)} + \frac{(\sin W) \sin W}{(\cos W + 1) \sin W} =$$

$$\frac{\cos^2 W + \cos W}{\sin W (\cos W + 1)} + \frac{\sin^2 W}{(\cos W + 1) \sin W} =$$

$$\frac{\sin^2 W + \cos^2 W + \cos W}{\sin W (\cos W + 1)} =$$

$$\frac{1 + \cos W}{\sin W (\cos W + 1)} =$$

$$\frac{1}{\sin W} =$$


$$(14) \frac{1}{1-\cot A} + \frac{1}{1-\tan A} = 1$$

$$\frac{(1-\tan A)}{(1-\tan A)(1-\cot A)} + \frac{(1-\cot A)}{(1-\tan A)(1-\cot A)} =$$

$$\frac{1-\tan A}{(1-\tan A)(1-\cot A)} + \frac{1-\cot A}{(1-\tan A)(1-\cot A)} =$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+\tan A \cot A} =$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+\tan A \frac{1}{\tan A}} =$$

$$\frac{2-\tan A-\cot A}{1-\cot A-\tan A+1} =$$

$$\frac{2-\tan A-\cot A}{2-\cot A-\tan A} =$$

$$1 =$$



$$(15) \frac{1}{1-\sin x} + \frac{1}{1+\sin x} = 2\sec^2 x$$

$$\frac{1}{(1-\sin x)(1+\sin x)} + \frac{1}{(1+\sin x)(1-\sin x)} =$$

$$\frac{1+\sin x}{(1-\sin x)(1+\sin x)} + \frac{1-\sin x}{(1+\sin x)(1-\sin x)} =$$

$$\frac{2}{1+\sin x-\sin x-\sin^2 x} =$$

$$\frac{2}{1-\sin^2 x} =$$

$$\frac{2}{\sin^2 x + \cos^2 x - \sin^2 x} =$$

$$\frac{2}{\cos^2 x} =$$

$$2\left(\frac{1}{\cos^2 x}\right) =$$

$$2\sec^2 x =$$

